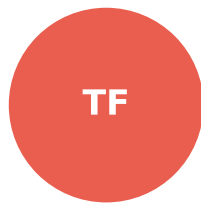


ATAC-seq Regulatory Analysis Pipeline

HCC Multiome | TC Hepatocytes | Cirrhosis vs Hepatitis

Causal Evidence Chain

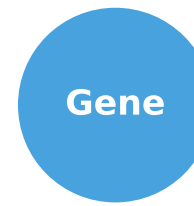
TF → DAR → Gene (3-node regulatory path)



binds



regulates



79 TFs

102 DARs

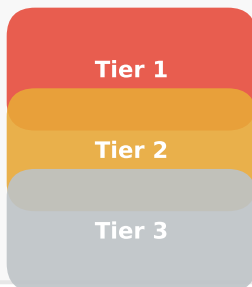
109 genes

5 Evidence Layers

- DA Significance: $-\log_{10}(\text{padj}) / 30$
- DA Effect Size: $|\log_2\text{FC}| / 5$
- Motif PWM Score: PWM score / 20
- Peak-Gen~~e~~ Linkage: $|z\text{-score}| / 10$
- TF-Gen~~e~~ Correlation: $|\text{Spearman } \rho| / 0.4$

Composite = $(ev_1 \times ev_2 \times ev_3 \times ev_4 \times ev_5)^{(1/5)}$

Evidence Tier Classification



All 5 evidence layers > 0

419 (1.8%)

PWM + Linkage + DA > 0

17,200 (73.0%)

DA + Linkage > 0

5,952 (25.2%)

Top 10 Tier 1 Master Regulators

Rank	TF	Targets	DARs	Score
1	HNF4A	55	43	0.301
2	NR1H2::RXRA	31	18	0.248
3	RXRG	28	23	0.229
4	NR2F1	22	19	0.260
5	RXRB	22	16	0.221
6	HNF4G	21	17	0.321
7	CTCF	19	17	0.306
8	HNF1B	16	18	0.277
9	TEAD4	13	10	0.255
10	HNF1A	11	14	0.251

Key Finding

HNF4A

Hepatocyte Nuclear Factor 4 Alpha

#1 Master Regulator of chromatin remodeling

in Cirrhosis-associated Hepatocytes

Notable HNF4A Target Genes

CYP3A4

Drug metabolism (cytochrome P450)

AGT

Angiotensinogen — blood pressure regulation

GPT

ALT enzyme — liver function marker

SEC14L2

Lipid transport & cholesterol metabolism

TF

Transferrin — iron transport

POR

P450 oxidoreductase

SLC27A5

Bile acid metabolism

55

target genes

43

mediating DARs

0.301

causal score